MODULE 02

Types of Data & Collection

Know Your Data Before You Analyze It

■ DATA FUNDAMENTALS

■ LEARNING OUTCOME

By the end of this PDF, you will classify different types of data, understand data sources and collection methods, and know how data flows from the real world into your analysis.

■ Skills You Will Gain
• Classify data by type (quantitative, qualitative)
• Understand structured vs unstructured data
• Know primary vs secondary data sources
• Apply correct data collection methods
• Understand data formats: CSV, JSON, XML, SQL

■ **Understanding data types prevents costly analysis mistakes. Wrong data type = wrong conclusions.**

SECTION 1

Data Type Classification

Category	Types & Examples
Quantitative (Numbers)	Discrete: Count of orders (3, 7, 12) Continuous: Temperature (36.7°C), Revenue (■1,23,456.78)
Qualitative (Categories)	Nominal: Colors, Country, Gender (no order) Ordinal: Ratings (1–5 stars), Education Level (has order)
Time Series	Data collected over time: Stock prices, Weather readings
Spatial / Geo	Location-based: GPS coordinates, City-level sales maps

SECTION 2

Structured vs Unstructured Data

Type	Description	Examples
Structured	Organized in rows & columns, easy to query	SQL tables, Excel sheets, CSV files
Semi-Structured	Has some organization but not rigid	JSON, XML, HTML, emails with headers
Unstructured	No fixed format, hardest to analyze	Images, videos, social media posts, PDFs

SECTION 3

Primary vs Secondary Data

Source	Definition	Methods
Primary	Data you collect yourself for your specific purpose	Surveys, interviews, experiments, sensors
Secondary	Data collected by others, reused for your purpose	Government data, Kaggle datasets, web scraping

SECTION 4

Data Collection Methods

1 **Surveys & Forms**
Google Forms, Typeform — best for opinions, ratings, demographics.

2 **Web Scraping**
Python (BeautifulSoup, Scrapy) — extract data from websites automatically.

3	**APIs**	REST APIs return structured JSON: weather, social media, finance data.
4	**Sensors / IoT**	Temperature sensors, wearables, traffic cameras — real-time streams.
5	**Database Queries**	SQL SELECT statements to pull records from business databases.

SECTION 5

Data Formats

CSV (Comma Separated Values)

```
name,age,salary
Alice,28,50000
Bob,32,65000
```

JSON (JavaScript Object Notation)

```
{"name": "Alice", "age": 28, "salary": 50000}
```

SQL Table

```
SELECT name, salary FROM employees WHERE age > 25;
```

SECTION 6

Data Quality Dimensions

Dimension	Question to Ask
Completeness	Are there missing values? Are all required fields filled?
Accuracy	Is the data correct and error-free?
Consistency	Is the same data stored the same way everywhere?
Timeliness	Is the data recent enough for the analysis?
Validity	Does the data match the expected format/range?
Uniqueness	Are there duplicate records?

■ PRACTICE QUESTIONS

Q1. Is 'Customer Rating (1–5)' quantitative or qualitative?

> ***Answer:*** *Qualitative Ordinal — it has a meaningful order but is not a true numeric measurement.*

Q2. What is the difference between structured and unstructured data?

> ***Answer:*** *Structured fits in rows/columns (SQL tables); unstructured has no fixed format (images, text).*

Q3. Name two ways to collect primary data.

> ***Answer:*** *Surveys and direct observation/experiments.*

Q4. What format does most APIs return data in?

> ***Answer:*** *JSON (JavaScript Object Notation).*

Q5. What does data 'completeness' mean?

Answer: *All required fields are present with no missing values.*

■ MINI PROJECT

Apply what you've learned with a real dataset!

Dataset: Student Survey CSV: Name, Age, Study_Hours, Exam_Score, Preferred_Subject

Steps:

1	**Identify** Which columns are numeric? Which are categorical?
2	**Check** Any missing values? Any duplicates?
3	**Summarize** Average study hours, most popular subject
4	**Visualize** Pie chart: Subject preference distribution
5	**Insight** Is there a link between study hours and exam score?

Expected Output: Cleaned dataset + summary statistics + one chart

Insights to Find: Subject popularity, average performance, data quality issues found

■ Data Types Quick Reference	
Discrete	Whole numbers — count of items
Continuous	Any value — height, weight, revenue
Nominal	Categories, no order — colors, country
Ordinal	Categories with order — ratings, grades
Structured	Rows & columns — SQL, CSV, Excel
Unstructured	No format — images, video, text
Primary	You collected it — surveys, experiments
Secondary	Someone else collected — Kaggle, govt
JSON	{ key: value } — most common API format
CSV	Comma-separated — simplest data format